



PLATO

RETHINKING JIT OPTIMIZATIONS: PRODUCTIZING HYBRID ANALYSIS WITH JITSERVER

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JIT Optimizations

- JIT runs **inside the JVM**, sharing resources with the application
- JIT **shares resources** with the JVM and application
- Multi-stage model: optimize hot methods

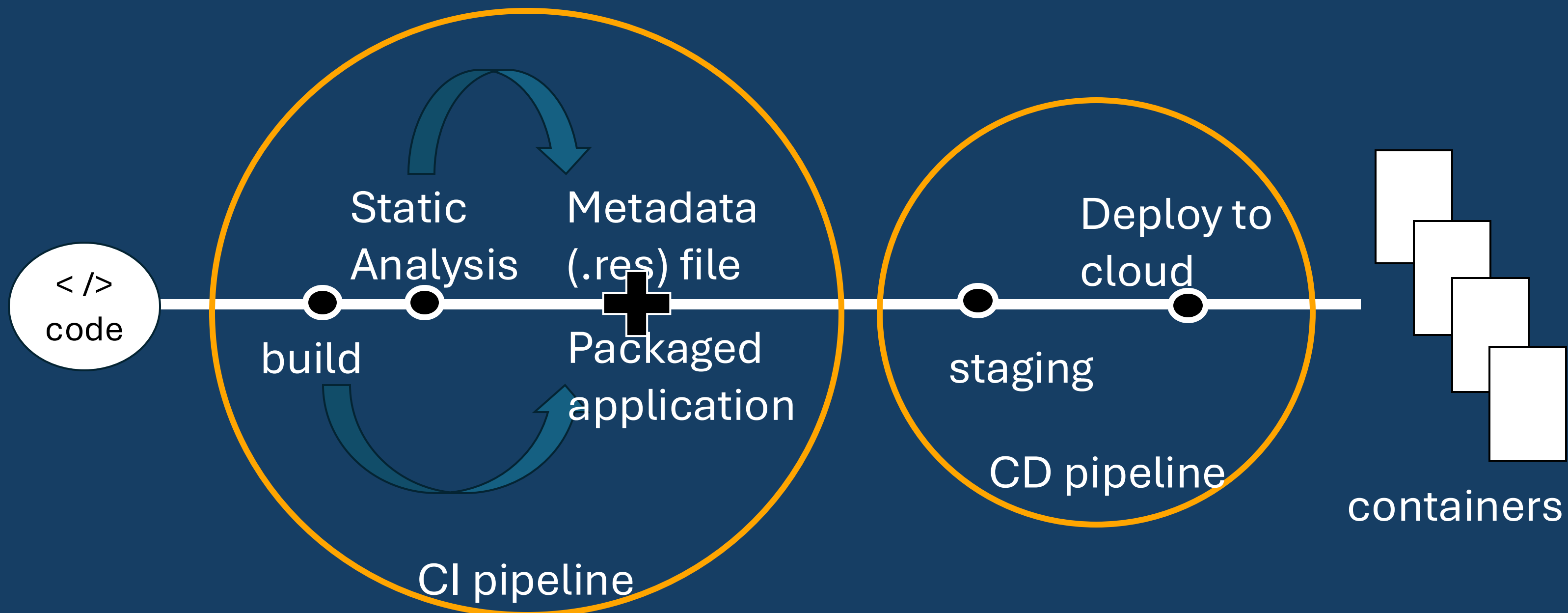
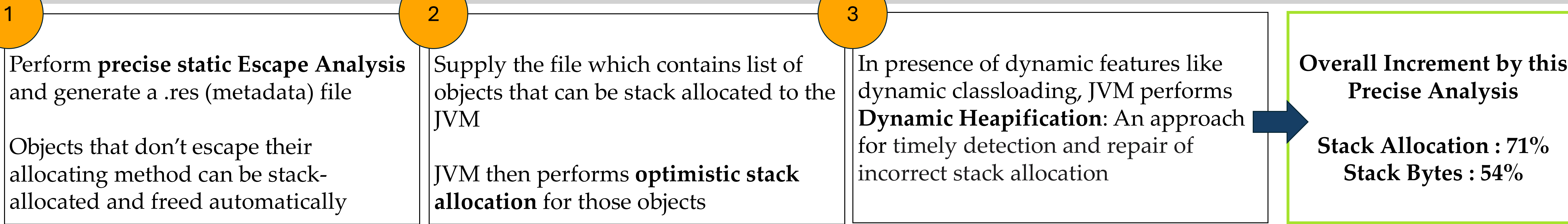
The Balancing Act

- JIT Compilation is a **game of tradeoff**
- Longer compilation time and higher memory usage versus faster runtime performance from better optimizations
- JITs often choose **conservative strategies**

Imprecise Analysis

- Strict **time/memory/compute budgets**
- Advanced optimizations (escape analysis, inlining, register allocation) not an option
- JITs prefer simplified analyses which is **less precise, lost performance potential**

Hybrid : Static + Dynamic Program Analysis [1]



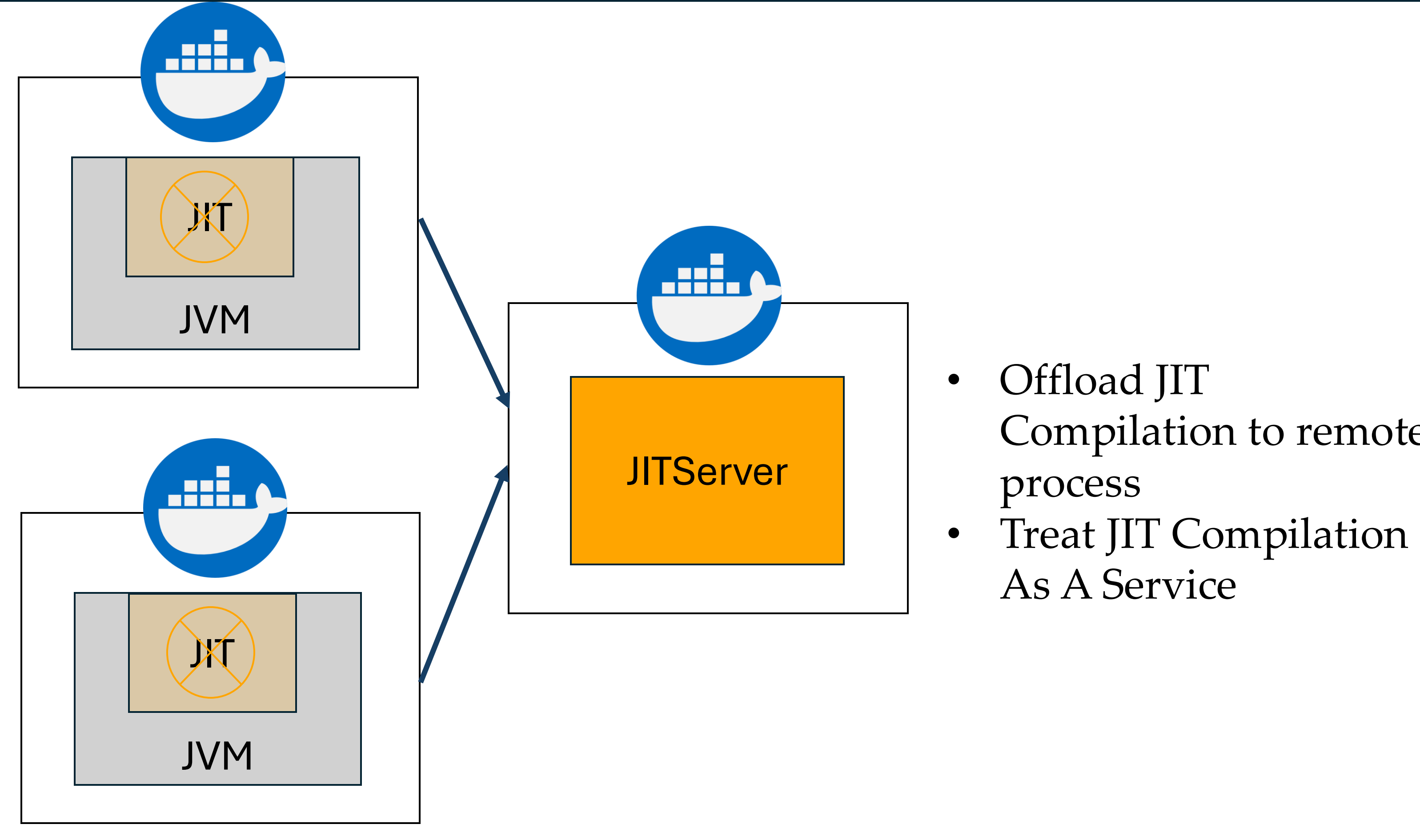
Deployment Issue

- **Extra build step**: CI/CD must generate .res files
- Clients may distrust **third-party** static analysis metadata and be wary of **trust, debugging, and maintainability** issues

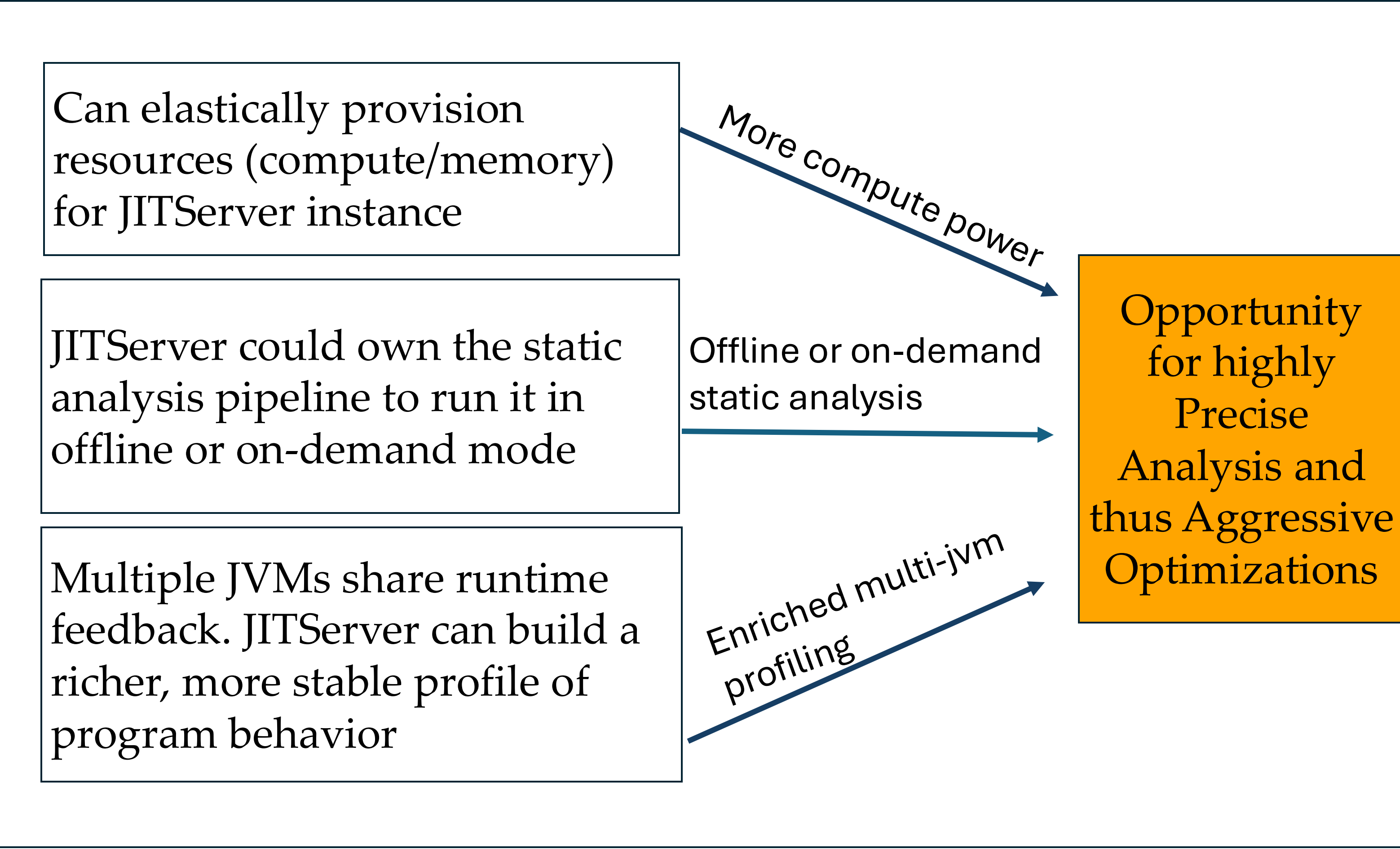
Overprovisioning

- Local JIT causes memory and CPU **spikes** during startup
- In order to avoid out-of-memory killing, **containers must over-provision** under constrained environments

JITServer : Disaggregated JIT Compiler [2]



Future Work



[1] : Aditya Anand, Solai Adithya, Swapnil Rustagi, Priyam Seth, Vijay Sundaresan, Daryl Maier, V. Krishna Nandivada, and Manas Thakur. 2024. Optimistic Stack Allocation and Dynamic Heapification for Managed Runtimes. Proc. ACM Program. Lang. 8, PLDI, Article 159 (June 2024)

[2] : <https://eclipse.dev/openj9/docs/jitserver/>